



Dhruv <dhruv.13datta@gmail.com>

IEEE Chandigarh Subsection International Conference : Submission (1896) has been created.

1 message

Microsoft CMT <noreply@msr-cmt.org>
To: dhruv.13datta@gmail.com

11 May 2026 at 22:10

Hello,

The following submission has been created.

Track Name: Artificial Intelligence and Data Science

Paper ID: 1896

Paper Title: An Intelligent Multi-Agent Framework for Enhanced Crop Yield Prediction and Agricultural Decision Support

Abstract:

This paper presents an intelligent five-agent framework for crop yield prediction and agricultural decision support. The system employs five specialized cooperative agents – Data Collection, Preprocessing, Learning, Decision-Making, and Feedback – coordinated through a shared Knowledge Base. Evaluated on 25,932 deduplicated FAO records spanning 101 countries, 10 crop types, and the period 1990–2013, the Agent-Based Random Forest model achieved an R^2 of 0.986, MAE of 4,078 hg/ha, RMSE of 10,252 hg/ha, and 84.6% prediction accuracy ($\pm 15\%$ tolerance), substantially outperforming Linear Regression ($R^2 = 0.745$, Accuracy: 21.3%) and LinearSVR baselines. The preprocessing pipeline incorporates one-hot encoding, kNN imputation, Min-Max normalization, and variance-based feature selection across 114 features. An adaptive Feedback Agent enables iterative retraining, ensuring robustness across diverse agro-climatic conditions. The Decision-Making Agent translates yield forecasts into actionable irrigation, fertilization, and harvesting advisories across three confidence tiers, achieving an 84.8% high-confidence classification rate. These results validate the multi-agent paradigm as an effective, modular, and interpretable approach for end-to-end agricultural AI pipelines, establishing a strong baseline for future integration of satellite imagery, IoT sensor data, and deep temporal forecasting models.

Keywords: Multi-agent systems; crop yield prediction; Random Forest; precision agriculture; decision support; FAO dataset; machine learning

Created on: Mon, 11 May 2026 16:39:54 GMT

Last Modified: Mon, 11 May 2026 16:39:54 GMT

Authors:

- dhruv.13datta@gmail.com (Primary)

Secondary Subject Areas: Not Entered

Submission Files:

Multiagein crop yield prediction.pdf (675 Kb, Mon, 11 May 2026 16:38:10 GMT)

Submission Questions Response:

1. Corresponding author Email ID
dhruvdatta11@gmail.com,
AbhinavBhar2004@gmail.com,
DivanshuKaushik123@gmail.com
gurpreet.1309@chitkara.edu.in
2. Contact number with country code
+91 9914141580
3. Author's Information
Agreement accepted
4. Co-author's Consent
Agreement accepted

11/05/2026, 22:15

Gmail - IEEE Chandigarh Subsection International Conference : Submission (1896) has been created.

5. Certificate of Originality

Agreement accepted

6. Author's Consent

Agreement accepted

7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

Agreement accepted

Thanks,
CMT team.

Please do not reply to this email as it was generated from an email account that is not monitored.

To stop receiving conference emails, you can check the 'Do not send me conference email' box from your User Profile.

Microsoft respects your privacy. To learn more, please read our [Privacy Statement](#).

Microsoft Corporation
One [Microsoft Way](#)
[Redmond, WA 98052](#)